

Algebra II Sequences and Series Cheatsheet

Module 8

Sequence vs Series

sequence: list of terms

series: sum of terms

Arithmetic Sequences

$$a_n = a_1 + (n - 1)d$$

$$d = a_n - a_{n-1}$$

Recursive form:

$$a_n = a_{n-1} + d, \quad a_1 = \text{first term}$$

Arithmetic Series

$$S_n = \frac{n(a_1 + a_n)}{2}$$

$$S_n = \frac{n}{2} (2a_1 + (n - 1)d)$$

Geometric Sequences

$$a_n = a_1 r^{n-1}$$

$$r = \frac{a_n}{a_{n-1}}$$

Recursive form:

$$a_n = r a_{n-1}, \quad a_1 = \text{first term}$$

Geometric Series

$$S_n = \frac{a_1(1 - r^n)}{1 - r} \quad (r \neq 1)$$

Infinite geometric series:

$$S_\infty = \frac{a_1}{1 - r} \quad (|r| < 1)$$

If $|r| \geq 1$, the infinite geometric series diverges.

Sigma Notation

$$\sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$$

$$\sum_{k=1}^n c = cn$$

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$$

Quick Checks

- Arithmetic means constant difference.
- Geometric means constant ratio.
- Infinite geometric sums require $|r| < 1$.
- Match the starting index to the first term.

Common Sigma Formulas

For a positive integer n :

$$\sum_{k=1}^n k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

For consecutive integers beginning at m instead of 1, subtract the terms before m :

$$\sum_{k=m}^n k = \sum_{k=1}^n k - \sum_{k=1}^{m-1} k = \frac{n(n+1) - (m-1)m}{2}.$$

The same subtraction idea works for sums of squares and cubes.

Sequence and Series Comparison Table

Let $S_n = \sum_{k=1}^n a_k$. The last two columns refer to the series formed by adding the sequence terms. Assume $a_1 \neq 0$ in the geometric row unless noted.

Function / sequence name	Recursive form	Explicit form	$\lim_{n \rightarrow \infty} a_n$	Partial sum S_n	Infinite sum $\sum_{n=1}^{\infty} a_n$
Constant sequence	$a_1 = c; a_n = a_{n-1}$	$a_n = c$	c	cn	0 if $c = 0$; otherwise diverges
Consecutive integers	$a_1 = 1; a_n = a_{n-1} + 1$	$a_n = n$	$+\infty$	$\frac{n(n+1)}{2}$	Diverges
Arithmetic sequence	a_1 given; $a_n = a_{n-1} + d$	$a_n = a_1 + (n-1)d$	$+\infty$ if $d > 0$; a_1 if $d = 0$; $-\infty$ if $d < 0$	$\frac{n}{2}(2a_1 + (n-1)d)$	Diverges unless $a_1 = d = 0$
Square numbers	$a_1 = 1; a_n = a_{n-1} + 2n - 1$	$a_n = n^2$	$+\infty$	$\frac{n(n+1)(2n+1)}{6}$	Diverges
Cube numbers	$a_1 = 1;$ $a_n = a_{n-1} + 3n^2 - 3n + 1$	$a_n = n^3$	$+\infty$	$\left(\frac{n(n+1)}{2}\right)^2$	Diverges
Geometric sequence	a_1 given; $a_n = ra_{n-1}$	$a_n = a_1r^{n-1}$	0 if $ r < 1$; a_1 if $r = 1$; otherwise depends on r or does not exist	$\frac{a_1(1-r^n)}{1-r}$ if $r \neq 1$; na_1 if $r = 1$	$\frac{a_1}{1-r}$ if $ r < 1$; otherwise diverges

Convergence check: If $\sum a_n$ converges, then $a_n \rightarrow 0$. A sequence approaching 0 is necessary for an infinite sum to converge, but it is not always sufficient.